

RESEARCH ARTICLE

IoT-Enabled Real-Time Monitoring and Disease Prediction System for Sustainable Aquaculture

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ABSTRACT The global shrimp aquaculture industry is highly vulnerable to water quality fluctuations, disease outbreaks and the inefficiencies of manual monitoring. To address these challenges, this paper presents a unified Artificial Intelligence (AI) and Internet of Things (IoT) enabled framework that integrates real-time sensing, predictive analytics and automated pH sensor calibration for precision aquaculture management. A custom-designed aqua buoy system continuously measures the critical parameters including pH, Dissolved Oxygen (DO) and Oxidation Reduction Potential (ORP) using a distributed IoT sensor network with cloud connectivity. A responsive web dashboard and mobile application deliver real-time visualization, farmer advisories and automated alert notifications, facilitating timely interventions. Machine learning models are employed in the proposed model for disease classification and intelligent decision-making: Random Forest achieves 98.35% accuracy in disease prediction, while Random Forest regression provides highly reliable pH voltage prediction by minimizing manual calibration requirements. Furthermore, Long Short-Term Memory (LSTM) networks are utilized for time-series forecasting, achieving up to 94% accuracy in predicting pH, DO and ORP trends, thereby enabling early corrective actions. The proposed method enhances monitoring efficiency, reduces shrimp mortality risk and supports environmentally sustainable farming practices. This integrated approach demonstrates the potential of combining IoT, disease classification & prediction and automated calibration techniques to transform aquaculture into a data-driven, precision-managed and resource-efficient industry.

INDEX TERMS Aqua bouy, machine learning algorithms, water quality monitoring, sensor calibration, precision aquaculture.

I. INTRODUCTION

Aquaculture, also referred to as aquafarming, is a vital practice involving the controlled cultivation of aquatic organisms such as fish, shrimp and aquatic plants. It plays a significant role in global food production, addressing the increasing demand for seafood while alleviating pressure on wild fish stocks. Conducted across diverse systems including ponds, tanks and oceanic cages aquaculture contributes directly to food security, particularly in regions experiencing rapid population growth and declining marine resources [1]. Beyond

its contribution to food production, aquaculture holds considerable economic, environmental and social importance. It creates employment opportunities, fosters income diversification and stimulates trade and investment in coastal and rural areas. These multifaceted contributions highlight the need to ensure its sustainability through effective monitoring and management practices.

In recent years, innovations in water quality monitoring have transformed aquaculture management. By integrating real-time sensor networks with advanced computational methods, including machine learning operators, can continuously monitor water quality parameters and detect anomalies before they escalate into major threats [2], [3].

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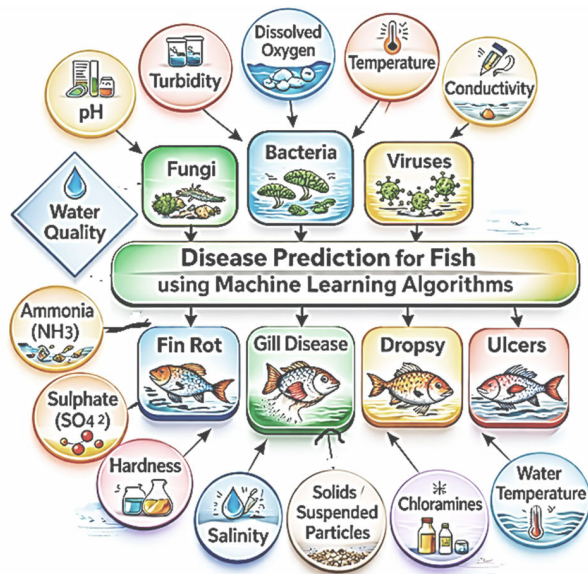


FIGURE 1. Interrelationship between water quality parameters and fish diseases.

Machine Learning (ML) enabled systems can process data from multiple monitoring points simultaneously, facilitating early detection of water quality deviations and supporting timely interventions to maintain optimal aquaculture conditions.

The fishbone diagram shown in Figure 1. illustrates the interrelationship between water quality parameters, machine learning algorithms and fish diseases. ML algorithms have proven particularly effective in disease prediction within aquaculture by analyzing complex datasets containing environmental and biological variables [4]. Such predictive models enable early identification of disease risks and support optimized disease management strategies, reducing losses and ensuring the long-term productivity of aquaculture operations. In shrimp farming, maintaining optimal water quality parameters is crucial for ensuring the health and growth of shrimp [5]. Parameters such as temperature, salinity, pH, ammonia, Dissolved Oxygen (DO) and nitrite levels directly affect shrimp physiology. Proper temperature ensures efficient metabolic activity, while salinity supports osmoregulation. pH stability is essential for enzymatic activity and overall metabolic functioning [6], [7]. Elevated ammonia or nitrite concentrations can be toxic, leading to stress and mortality. Adequate DO is vital for respiration and its depletion can cause reduced growth rates and higher susceptibility to disease. Together, these parameters create a balanced environment that promotes healthy and productive shrimp farming.

Several common shrimp diseases are closely linked to fluctuations in water quality parameters. White Spot Syndrome Virus (WSSV) and Early Mortality Syndrome (EMS) are frequently associated with elevated ammonia lev-

els and fluctuating temperatures [8]. Taura Syndrome Virus (TSV) outbreaks may occur when salinity and temperature deviate from optimal ranges for the farmed shrimp species. The prevalence of Infectious Hypodermal and Hematopoietic Necrosis Virus (IHHNV) is influenced by water temperature, salinity and stocking density, while Monodon Baculovirus (MBV) can also be linked to suboptimal water quality conditions [9]. For WSSV, water temperature should be maintained at 25–30 °C, ammonia levels below 0.02 mg/L, pH between 7–8, nitrite below 0.1 mg/L and DO above 5 mg/L, with salinity ideally below 20 ppt. EMS management requires similar ranges, with salinity adaptable to both low- and high-salinity environments. MBV control involves maintaining ammonia below 0.02 mg/L and DO above 5 mg/L, with nitrite levels between 0.1–0.3 mg/L. TSV management requires brackish to full-strength seawater (25–35 ppt), whereas IHHNV is generally associated with higher salinity environments and requires similar water quality control [10]. Maintaining these optimal ranges significantly minimizes the risk of disease outbreaks and supports a stable production environment. The pH measurement plays a critical role in aquaculture and other water-related applications. However, pH sensors are prone to signal drift due to long-term use and environmental factors, which can lead to erroneous data and poor decision-making. Regular calibration is essential to maintain data accuracy. Traditional calibration methods involve manual adjustments against known reference standards, which is challenging in real-time applications. To address this limitation, an automated calibration approach can be implemented by establishing a relationship pH value in 5 to 12 range and corresponding voltage outputs [11]. This allows the development of a predictive model capable of estimating voltage values based on pH levels, ensuring accurate and reliable measurements for real-time monitoring systems.

Recent advances in water quality monitoring emphasize the integration of real-time sensors with artificial intelligence to improve the precision and responsiveness of data acquisition and interpretation. The use of smart sensor systems for real-time monitoring in brackish water environments and reported improved reliability and data precision. Similarly, the relevance of Generative Pre-trained Transformer (GPT) models in predictive analytics for aquatic systems showcases their potential for accurately forecasting water quality parameters. IoT-based systems for brackish water monitoring, such as simplified sequential data collection and enable remote supervision [12]. The ML techniques for predictive water quality monitoring and found Random Forest (RF) and Support Vector Machines (SVM) particularly effective for handling complex environmental data [13]. The improvements in disease prediction and resource management by integrating GPT models into aquaculture predictive frameworks. In literature the authors, emphasized the value of ML models for predicting water quality indices and enabling preventive interventions, while along

with the focus on IoT-enabled smart aquaculture systems for improving management practices [14]. Singh [15] highlighted the importance of strategic sensor placement in brackish water systems to ensure comprehensive data coverage. The cost effectiveness of low-cost IoT sensors for real-time water quality monitoring suggests their potential for widespread application. Collectively, these studies demonstrate how the combination of IoT, AI and predictive modeling enable multi-parameter sensing, remote monitoring via mobile applications and proactive decision-making for sustainable aquaculture operations.

This paper presents a comprehensive framework to enhance precision management in shrimp aquaculture by combining IoT sensing, automated analysis, calibration and real-time visualization. At the core of the proposed solution is the AquaBuoy module, a floating monitoring system equipped with multiple sensors that continuously measure key water quality parameters such as pH, DO and Oxidation Reduction Potential (ORP). Data from the sensors is transmitted using Message Queuing Telemetry Transport (MQTT), a lightweight publish-subscribe protocol optimized for IoT applications, enabling seamless and low-latency communication with the central server. The collected sensor data is then analyzed to perform disease prediction, identifying conditions that indicate a higher risk of shrimp disease outbreaks. This enables timely preventive actions and reduces the likelihood of stock losses. In addition, the system is designed to predict upcoming trends in water quality, supporting proactive pond management. To ensure accuracy, an automated pH calibration system is implemented by establishing the relationship between pH levels and their corresponding voltage outputs. This predictive approach eliminates the need for frequent manual calibration, maintains consistent sensor performance and ensures reliable data for decision-making.

Finally, all processed data, predictions and calibration results are displayed through a mobile application and dashboard, providing users with real-time visualization, alerts and recommendations. Farmers can remotely monitor their ponds, respond to potential issues quickly and maintain optimal conditions efficiently. By integrating Aquabouy sensing, sensor data analysis, disease prediction, automated calibration and user-friendly visualization, this system transforms aquaculture management into a data-driven process that improves monitoring efficiency, minimizes risks and supports sustainable and scalable shrimp farming operations. Finally, all processed data, predictions and calibrations results are displayed through a mobile application and dashboard, which provides the data analysis, disease prediction, automated calibration and user-friendly visualization.

The scientific contributions of the proposed work presents an integrated aquaculture monitoring and disease prediction framework that combines real-time water quality data acquisition, edge-based intelligence, and scientific modeling of disease-environment interactions. Calibrated sensors

mounted on an IP68-rated AquaBuoy platform continuously monitor key parameters such as pH, dissolved oxygen, and temperature, ensuring stable immersion and reliable real-time data under varying pond conditions. The system enables timely capture of rapid environmental changes through continuous wireless data transmission. Lightweight TinyML models are deployed directly on edge devices to perform low-latency, energy-efficient prognostics using locally processed sensor data. These models are trained on historical and multi-site datasets to enhance robustness and generalization for early detection of stress and disease conditions. Furthermore, a water quality-disease relationship model is developed using established physiological thresholds, statistical correlation, time-series analysis, and feature importance evaluation. This integrated approach provides scientifically grounded, adaptive, and proactive decision support for sustainable aquaculture management.

The remainder of this paper is structured as follows: Section II provides a comprehensive review of existing literature, establishing the context for the study. Section III details the proposed methodologies and techniques. Section IV presents and analyzes the experimental results, followed by a discussion of the key findings. Finally, Section V concludes the work by summarizing the major contributions, outlining the implications of the study and suggesting potential directions for future research.

II. LITERATURE REVIEW

The integration of Internet of Things (IoT) devices and Artificial Intelligence (AI) in aquaculture has significantly enhanced real-time monitoring, predictive analysis, and overall farm management. AquaBuoy platforms enable continuous remote monitoring of critical water quality parameters such as temperature, pH, dissolved oxygen (DO), turbidity, and salinity, thereby facilitating early detection of stress conditions and hypoxia in aquaculture ponds. UAV-assisted AquaBuoy systems have further improved spatial coverage and measurement accuracy, particularly in large water bodies; however, challenges related to battery life, calibration, and maintenance remain [16]. Long-term trend analysis supported by these platforms enables optimization of feeding strategies, oxygen management, and nutrient balance, contributing to improved farm productivity.

Sensors form the backbone of water quality monitoring systems, with multi-parameter wireless sensors capable of measuring pH, total dissolved solids (TDS), conductivity, DO, turbidity, and heavy metals, providing accurate real-time data for proactive aquaculture management. Low-cost sensors have demonstrated reliable performance even in brackish water environments, enabling adoption by small- and medium-scale farmers [17]. Strategic sensor placement, often complemented by UAV-assisted monitoring, improves spatial assessment of water quality, although sensor drift, calibration, and periodic maintenance continue to pose challenges.

Machine Learning (ML) algorithms play a crucial role in interpreting sensor data for water quality prediction and disease detection. Random Forest models have achieved high prediction accuracy (94.16%) for multiple water quality parameters [18], while Support Vector Machines (SVM) and Artificial Neural Networks (ANN) effectively model non-linear relationships and temporal variations in environmental data. These approaches not only enhance predictive accuracy but also support proactive decision-making, thereby reducing operational costs and improving farm efficiency. In addition, voltage prediction and pH sensor calibration are critical for reliable water quality assessment. AI-driven calibration techniques, including neural networks and SVM-based models, have outperformed traditional methods by compensating for sensor drift, non-linearity and environmental variability, enabling real-time correction and improved data reliability [19].

Visualization and decision-support tools play an essential role in translating sensor data and ML predictions into actionable insights. Smart dashboards enable real-time visualization of key parameters and support predictive analytics for trend monitoring and disease alerts. Mobile applications extend these capabilities by providing remote access and real-time notifications for abnormal water quality conditions and disease risks related to pH variations and feeding schedules, thereby supporting timely interventions [20]. Although recent studies have explored the use of GPT and AI-driven predictive models for trend forecasting, the integrated use of real-time sensing, advanced ML algorithms, automated calibration, and decision-support systems remains an emerging research area requiring further empirical validation.

Despite significant advancements in IoT-enabled sensors [21], UAV-based water quality assessment, and ML-based predictive analytics, several research gaps persist. Existing approaches often focus on isolated components such as monitoring, sensor calibration, or disease prediction, without integrating these elements into a unified, end-to-end system [22]. Traditional calibration techniques remain susceptible to environmental variability and sensor drift, while ML applications for real-time voltage prediction using pH values are limited in scope. Furthermore, many systems lack farmer-friendly deployment tools, including multilingual dashboards and alert-based mobile interfaces [23]. The comparative analysis presented in Table 1 highlights these gaps and provides the rationale for the proposed research framework, which integrates IoT-based sensing, automated calibration, machine learning-driven prediction, and a real-time decision-support dashboard to enable proactive, scalable, and sustainable shrimp aquaculture management. The next section details the architecture and components of the proposed system, emphasizing how it advances the state of the art.

The comparative analysis in Table 1 reveals several critical research gaps in current aquaculture monitoring systems.

TABLE 1. Methodology, findings and research gaps from the literature.

Technology Category	Author (s) / Year	Technology	Key Parameters	Findings/ Research Gap
IoT + ML for Water Quality Prediction & Automation	Shete et al., 2025 [24]	IoT-enabled ensemble ML model	Temperature, pH, DO, Water Quality Index	High predictive performance; limited generalizability across diverse aquaculture environments.
	Razali, 2025 [25]	IoT + Random Forest ML classifier	Temperature, turbidity, DO	Effective real-time classification, but lacks integration with broader decision support or disease prediction.
	Pinit Nuangpirom et al., 2025 [26]	Edge ML (ESP32) for real-time prediction	Temperature, pH, DO	Demonstrates low-cost edge deployment; future work needed on larger seasonal datasets and federated learning.
	Halim et al., 2024 [27]	IoT + multiscale feature fusion (Conv-AE + GRU)	Multiple water quality indicators	Provides deep learning forecasting; applicability limited to aquaponics rather than broader aquaculture.
IoT-centric Monitoring Systems (Non-ML or analytics pending)	Oluseye et al., 2024 [28]	IoT intelligent water quality system	Temperature, pH, turbidity	Good baseline IoT monitoring; lacks predictive analytics or ML integration.
	Jais. et.al [29]	Improved low-cost sensor monitoring	Temperature, pH, DO, turbidity	Focused on sensor accuracy; does not incorporate ML predictions or adaptive control.
	Madeo, D., Pozzetti et al 2020 [30]	Unmanned Surface Vehicle for pervasive water quality monitoring	Temperature, pH, DO, conductivity	Demonstrates real-time monitoring using Autonomous platforms; lacks integration with ML predictive analytics
	Akhter, F., 2022 [31]	Portable IoT water quality monitoring system with multifunctional sensor	Temperature, pH, DO, turbidity	Shows multifunctional sensors in IoT; research gap in aquaculture-specific deployment and integration with decision-making ML models
	Eso et al., 2024 [32]	IoT water quality system for shrimp farming	pH, temperature, DO	Basic monitoring without advanced modeling or predictive functions. (MDPI)

TABLE 1. (Continued) Methodology, findings and research gaps from the literature.

Dashboard & Interface Systems	Flores-Iwasaki et al., 2025 [33]	IoT sensor dashboards	Water quality parameter	UI standardization and unified analytics missing.
	Smith 2023 [34]	Smart dashboards	Temperature, pH, DO, turbidity	Visualization focus; deeper ML/automation features.
Signal Processing Focus & Auxiliary Monitoring	Lin, 2024 [17]	Wireless multi-sensor network	pH, TDS, conductivity, Al, Ba, Cr	Sensor maintenance and drift issues; calibration consistency needed.
	Khan, 2023 [35]	IoT remote data collection	Temperature, pH, DO, turbidity	Effective design; coverage and power issues in larger ponds.

Existing IoT-ML approaches demonstrate strong predictive performance but suffer from limited generalizability, lack of multi-pond scalability, and insufficient integration with disease prediction and decision-support frameworks. Many IoT-centric monitoring systems focus primarily on real-time sensing without incorporating predictive analytics, automated calibration, or adaptive control mechanisms.

Dashboard and interface solutions emphasize visualization but lack unified analytics, actionable intelligence, and farmer-centric deployment features. Furthermore, calibration and signal-processing studies address sensor drift in isolation, without real-time adaptive updating or seamless integration into end-to-end aquaculture systems. Remote and UAV-assisted monitoring improves spatial coverage but remains constrained by power, calibration, and operational complexity. Addressing these gaps, the proposed methodology introduces an integrated and scalable framework that combines AquaBuoy-based modular sensing, automated pH calibration, edge-enabled machine learning for water quality and disease prediction, lightweight communication, and a real-time decision-support dashboard. This holistic approach advances beyond existing works by unifying sensing, prediction, calibration, and visualization into a single, practical system for proactive and sustainable shrimp aquaculture management.

Existing studies in IoT and machine learning-based aquaculture monitoring have demonstrated significant advances in real-time water quality measurement, predictive analytics, and automation. IoT-ML approaches provide high predictive performance but often suffer from limited generalizability across diverse aquaculture environments or are restricted to specific setups such as aquaponics. Several IoT monitoring systems enable real-time data collection but lack integration with predictive disease modeling and adaptive decision

support. Dashboard and interface solutions emphasize visualization, yet fail to provide automation, multi-pond scalability, or predictive capabilities. Furthermore, calibration and signal processing methods face challenges with sensor drift and real-time adaptive updates, while auxiliary monitoring systems encounter power, connectivity, and coverage limitations in larger ponds.

The proposed framework introduces several technically novel contributions to aquaculture monitoring and disease prediction. Unlike existing approaches, it integrates IP68-rated AquaBuoy sensors for robust, continuous, and real-time water quality monitoring, ensuring stable immersion and consistent measurements across diverse pond conditions. The deployment of TinyML models directly on edge devices enables low-latency, site-specific disease prediction without relying on cloud connectivity, addressing limitations of generalizability and remote operation in prior studies. Furthermore, the development of a scientifically grounded water quality–disease relationship model allows quantitative linkage between environmental fluctuations and shrimp health, supporting predictive and adaptive management. Real-time signal preprocessing and automated calibration routines ensure data reliability over long-term deployments, while multi-pond wireless monitoring and user-friendly dashboards provide actionable insights for operational decision-making. Collectively, these elements create a scalable, integrated, and scientifically validated framework that fills critical gaps in precision aquaculture, making the system both innovative and practically deployable.

III. METHODOLOGY

The proposed method contributes to new technologies by uniquely integrating robust sensing, uncertainty-aware calibration, noise-resilient signal processing, and a TinyML-based prediction model. The prediction modeling explicitly relates water quality parameters to disease occurrence within a single, long-term field-deployable aquaculture monitoring framework, an aspect that is not by existing methods. Table 2 provides a comparative summary of key technological aspects between existing aquaculture monitoring methods and the proposed system, to emphasize its scientific contribution. While both approaches support multi-parameter water quality sensing and machine learning-based prediction, existing methods largely lack robustness for long-term field deployment, formal calibration procedures, and noise-resilient signal conditioning. In contrast, the proposed method incorporates IP68-rated sensing hardware designed for continuous operation in harsh aquaculture environments, along with structured calibration and uncertainty analysis to improve measurement reliability. Furthermore, unlike prior studies that treat water quality monitoring and disease assessment independently, the proposed framework explicitly models the relationship between water quality dynamics and disease occurrence. The integration of sensing, signal conditioning, edge-level machine learning, and disease-aware analytics into a single end-to-end monitoring

TABLE 2. Comparison of key technologies with existing studies.

Key Technology / Aspect	Existing Methods [24-35]	Proposed Method
Multi-parameter sensing (pH, DO, Temp, EC)	✓	✓
IP68-rated long-term field deployment	✗	✓
Formal calibration & uncertainty analysis	✗	✓
Noise-resilient signal conditioning	✗	✓
Machine learning-based prediction	✓	✓
Water-quality–disease relationship modeling	✗	✓
End-to-end integrated monitoring system	✗	✓

system distinguishes the proposed approach from existing methods and demonstrates its novelty beyond incremental improvements.

The design and implementation of the proposed smart aquaculture monitoring system, which integrates IoT-based sensing, real-time data communication, automated calibration, predictive analytics and visualization tools is proposed. The primary objective is to provide shrimp farmers with a comprehensive and scalable solution for proactive pond management. The proposed system is shown in Figure 2 (a). integrates IoT and machine learning techniques to provide an intelligent solution for aquaculture water quality monitoring and disease prediction. Sensors deployed in the pond through an Aquabouy device continuously measure parameters such as pH, which are then transmitted to a ESP 32. This processes the data and sends it via Wi-Fi module to a cloud database for storage and accessibility, while also displaying the readings on a local display unit for immediate reference. In the cloud, the collected data undergoes preprocessing to remove noise, normalize values and prepare it for further analysis. Feature selection techniques are then applied to extract the most relevant attributes, which are used to train and test machine learning algorithms for predictive modeling.

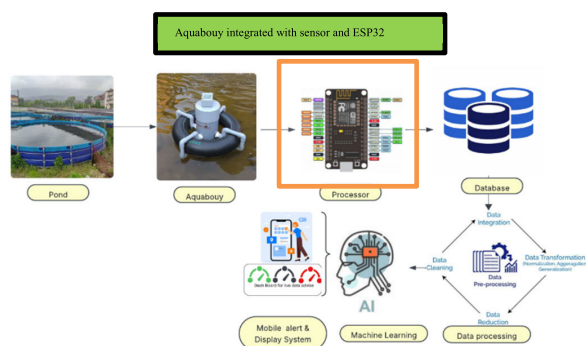
The system focuses on predicting voltage variations linked to pH levels, which helps in accurately identifying potential risks. Based on these predictions, the system performs disease forecasting to identify possible outbreaks caused by poor water quality. A user-friendly dashboard is designed to visualize real-time data and prediction outcomes, while an alert system provides timely notifications to farmers whenever abnormal conditions or disease risks are detected. This integrated framework ensures continuous monitoring, early disease prediction and proactive decision-making, thereby supporting sustainable aquaculture practices.

Figure 2 (b). presents the process flow diagram outlining the sequential stages of the proposed framework, including real-time data acquisition, cloud-based storage, machine learning model execution, and dashboard visualization for farmer decision-making. The process flow of the proposed method, depicting the integration of IoT sensors, the Aquabouy monitoring system, and the AI-driven predictive engine, culminating in actionable insights delivered

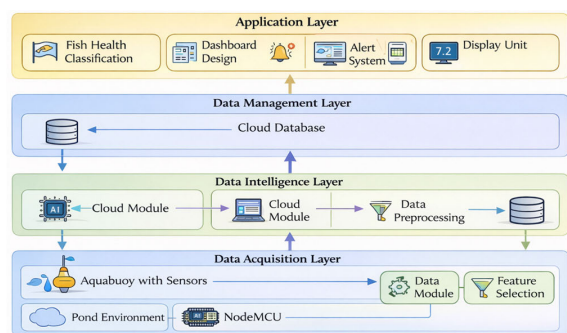
via the centralized dashboard and mobile application [34], [17]. The proposed Aquabouy module with integrated sensors is designed to provide comprehensive monitoring of aquaculture pond water quality. It integrates multiple sensors, including pH, DO, TDS, temperature and turbidity, which together offer critical insights into aquatic health. These parameters are vital indicators, as deviations can significantly impact fish or shrimp survival, growth and disease susceptibility. The sensor array is deployed within the pond to capture real-time measurements that form the foundation for further analysis.

At the core, the Aquabouy system has a microcontroller integrated with a Wi-Fi module for communication. This component acts as the primary processing and communication hub, interfacing directly with the sensors to collect raw data. Once acquired, the data is transmitted wirelessly to a cloud-based database for storage and subsequent processing. Simultaneously, the controller supports a local display unit, allowing farmers to view real-time water quality readings on-site. This dual capability ensures both instant feedback and long-term data management, enabling effective monitoring even in resource-constrained environments. To ensure reliable and efficient data transmission the system employs the MQTT protocol. This follows a publish subscribe model, where the controller functions as the publisher, transmitting sensor readings to an agent, while the cloud services and monitoring applications act as subscribers. The lightweight nature of MQTT, coupled with its support for multiple Quality of Service (QoS) levels, guarantees low-overhead, scalable and real-time communication. This makes it well suited for continuous aquaculture monitoring, even under unstable or low-bandwidth network conditions.

Beyond cloud-based analytics with on-device intelligence, the Aquabouy system incorporates Tiny Machine Learning (TinyML) to enable real-time operation of the systems. TinyML models and lightweight activation functions are deployed on the controller to process sensor data locally. This allows the system to perform anomaly detection such as sudden pH shifts, oxygen depletion, or turbidity spikes without requiring constant cloud connectivity. By embedding predictive capabilities at the edge, the module ensures faster response times, reduces network dependency and enhances overall system autonomy. The operational workflow begins with the integrated sensor array measuring water quality parameters in system, which are then processed and transmitted by the controller via Wi-Fi. The data is simultaneously sent to a local display for immediate access and to a cloud database through MQTT for storage and advanced analysis. In the cloud, the data undergoes preprocessing, noise removal and feature selection to prepare it for machine learning algorithms. Trained models are then deployed for disease prediction and trend analysis. Meanwhile, TinyML-enabled edge processing provides real-time predictions and early alerts to farmers. This hybrid cloud–edge architecture ensures robust monitoring, proactive decision-making and sustainable aquaculture practices.



(a). Real-Time Monitoring and Disease Prediction System



(b). Process Flow Diagram

FIGURE 2. Proposed IoT enabled real-time monitoring and disease prediction system architecture.

A. SENSORS WITH CALIBRATION AND MEASUREMENT UNCERTAINTY

The flexible pH sensor strips were constructed using screen printing technique. Electrochemical pH sensor in a two-electrode architecture was first designed length of the reference electrode and working electrode are maintained 1.7 cm and 1.5 cm respectively. The working area of the two-electrode system was maintained at 4 mm. Conductive graphene ink coating was utilized as the working electrode, and an Silver/Silver Chloride layer was used as the reference electrode in the two-electrode architecture on polyethylene terephthalate substrate. After this, the printed electrodes were cured at 60° for 60 min. An insulation layer was placed over the printed sensor layer for the hydrophobic features which is used to help the precise sample volume and to avoid the electrolyte flow to the contact pads. The pH measurements were performed using potentiometric method. Disodium hydrogen phosphate dihydrate, sodium hydroxide, hydrochloric acid, potassium hydrogen phthalate, potassium dihydrogen phosphate, Citric acid monohydrate, borax, and potassium chloride, potassium hydrogen phthalate was used to prepare various pH test solution.

A formal calibration procedure was followed for each water quality parameter and the corresponding details are provided in Table 3. All sensors were calibrated at the Indian Council of Agricultural Research – Central Institute of Brackishwater Aquaculture (ICAR-CIBA) laboratory

**FIGURE 3.** pH sensor calibration in different temperature accuracy = actual.**TABLE 3.** Sensor calibration details for water quality parameters [37].

Parameter	Sensor Type	Calibration Method	Buffer / Reference Solution	Applicable ISI / International Standard	Calibration Frequency
pH	Glass electrode pH sensor	Multi-point calibration (2- or 3-point)	pH 4.00, pH 7.00, pH 9.18 / 10.01 buffer solutions	IS 3025 (Part 11); ISO 10523: 2012	Monthly
Dissolved Oxygen (DO)	Electrochemical / Optical DO sensor	Zero-point + span calibration	Zero DO: Sodium sulfite (Na_2SO_3) solution; Span: Air-saturated water	ISO 5814: 2012	Monthly
Temperature	Digital temperature probe (NTC/P T100)	Single-point / comparison calibration	Traceable reference thermometer	IS 2848	Monthly

using higher-standard, certified reference equipment. The calibration frequency of the reference instruments was maintained in accordance with international standards. To ensure long-term accuracy of the proposed system, sensor outputs were periodically cross-validated and recalibrated against the certified reference instruments. The results demonstrate a close agreement of up to 99.9% with the standard calibrated equipment, confirming the reliability of the measured data.

To measure the pH, the printed electrodes were immersed in a corresponding pH solution (20 mL) and the measurements were performed for the duration of 20 seconds. The pH sensors value mainly depend on the operating temperatures, the buffers used to calibrate the pH sensors immersed on it

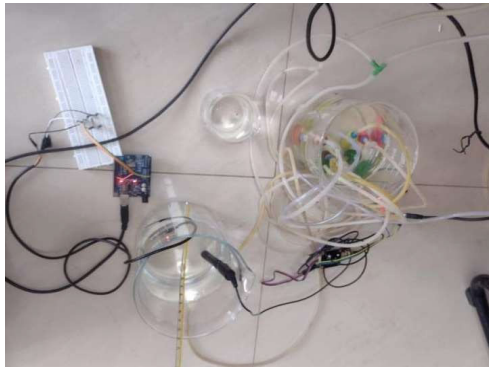


FIGURE 4. DO sensor calibration.

and base values are noted, the base values of pH sensor for different operating temperature are collected to calibrate the sensors as shown in Figure 3.

DO sensor Calibration in different salinity and temperature Accuracy = Actual DO $\pm$ 0.3. The commercial DO sensors in Figure 4 showcase the DO calibration and testing are insensitive to salt and brackish waters. The DO sensors are calibrated by increasing the salinity level. The DO is reduced with sodium sulfite (Na_2SO_3) and salinity is improved with Sodium Chloride (NaCl). Temperature sensors are calibrated with known mercury-based thermometers. The constant errors are observed and solved with constants. TDS sensors are calibrated with the known salinity values. Turbidity sensors which are commercially available are infrared based, which provide inaccurate results in real-time. The sensor is reconfigured with the visible white light and standardized with different Nephelometric Turbidity Units (NTU) solutions and calibrated.

Measurement uncertainty was quantified using traceable reference standards and manufacturer-specified sensor accuracy, in accordance with internationally accepted uncertainty evaluation practices. All reference instruments used for calibration were certified and traceable to national standards. Sensor accuracy and reference standard uncertainty were combined to estimate the expanded measurement uncertainty using a coverage factor of $k = 2$, corresponding to a confidence level of approximately 95%. The resulting uncertainty values for each water quality parameter are summarized in Table 4, demonstrating that the proposed sensing system operates within acceptable uncertainty limits for long-term aquaculture monitoring.

B. NOISE FILTERING AND SIGNAL CONDITIONING

Sensor measurements were affected by both environmental noise (temperature fluctuations, salinity variations) and electronic noise arising from sensor interfaces and analog signal transmission. Noise analysis was performed by observing short-term variations in sensor outputs under steady-state conditions prior to digital processing. To mitigate noise effects, analog signal conditioning was implemented using low-noise amplification to ensure compatibility with

TABLE 4. Sensors uncertainty and reference standard [37].

Sensor	Measurement Range	Sensor Accuracy	Reference Standard Uncertainty	Expanded Uncertainty ($k = 2$)
pH	0–14	± 0.02 pH	± 0.01 pH	± 0.04 pH
DO	0–20 mg/L	± 0.3 mg/L	± 0.1 mg/L	± 0.5 mg/L
Temperature	0–50 °C	± 0.1 °C	± 0.05 °C	± 0.2 °C
TDS	0–50 ppt	$\pm 2\%$	$\pm 1\%$	$\pm 4\%$
Turbidity	0–1000 NTU	$\pm 2\%$	$\pm 1\%$	$\pm 4\%$

the Analog to Digital Converter (ADC) of the microcontroller. Analog low-pass filtering was employed to suppress high-frequency disturbances before digitization.

Following ADC conversion, digital filtering techniques were applied to the acquired data. A moving-average filter was used to reduce random noise and smooth transient fluctuations while preserving the underlying temporal trends essential for water quality analysis and machine-learning-based prediction. The conditioned and filtered signals were subsequently processed by the microcontroller and transmitted to the cloud platform for storage, analysis, and visualization. This combined analog and digital filtering approach enhanced signal reliability and contributed to the overall robustness of the proposed system. The majority of sensors used are Modbus-based, while some are analog. Analog signals are converted via ADC and all sensor data undergoes preprocessing using a moving average filter to remove small anomalies and noise. Additionally, the data is periodically cross-validated against manual water quality measurements to ensure reliability and accuracy. These procedures collectively enhance the robustness and credibility of the sensor readings.

C. MACHINE LEARNING MODELS FOR DISEASE PREDICTION

Logistic regression statistical model used to estimate the probability of binary outcomes such as the presence or absence of a disease. The model computes a weighted sum of input variables and applies a sigmoid activation function to generate probabilities. Errors between predicted and actual outcomes are minimized using gradient descent and the process is iteratively repeated until convergence. The trained model outputs disease likelihood predictions, offering interpretability through feature coefficients that highlight the significance of each parameter. The multilayer perceptron a feedforward artificial neural network, captures nonlinear and complex relationships among input features. It consists of multiple layers of neurons, each applying activation functions to the weighted inputs. Training involves a forward pass, where outputs are computed and a backward pass, where errors are propagated to adjust weights using gradient

descent. This iterative optimization across multiple epochs enables the model to learn intricate mappings between water quality indicators and disease outcomes, thereby improving predictive accuracy. The Random Forest algorithm is an ensemble learning method that constructs multiple decision trees using bootstrap samples and random feature subsets. Each tree is trained independently and predictions are aggregated through majority voting (classification) or averaging (regression). By leveraging diverse subsets of data and features, Random Forest mitigates overfitting while ensuring high accuracy and robustness. It is particularly effective in capturing nonlinear interactions among aquaculture parameters. The Support Vector Machine (SVM) is a supervised learning algorithm that identifies an optimal hyperplane to maximize the margin between classes (disease vs. no disease). For nonlinear problems, kernel functions are employed to project input features into higher-dimensional spaces, enabling the identification of complex decision boundaries [34]. Once trained, the SVM classifies new instances based on their position relative to the hyperplane, providing precise discrimination in disease prediction tasks. The parameters employed for disease prediction encompass pH, turbidity, organic carbon, conductivity, sulfate, chloramines, solids and hardness levels. These indicators are critical determinants of water quality and directly influence the health and survivability of aquatic organisms. By integrating these features into machine learning models, the system can identify correlations between water quality variations and the likelihood of disease outbreaks.

In practice, Logistic Regression estimates probabilities based on linear feature combinations; MLP captures nonlinear associations through multiple hidden layers; Random Forest builds an ensemble of decision trees to generalize across diverse feature subsets; and SVM establishes optimal class separation through hyperplane maximization. Collectively, these approaches enable robust prediction of aquaculture diseases, thereby facilitating real-time decision support and sustainable fish farming practices. The effectiveness of machine learning models in predicting aquaculture diseases depends on their ability to handle diverse data, capture nonlinear relationships and provide interpretable outputs. The Logistic Regression, Multilayer Perceptron (MLP), Random Forest (RF) and Support Vector Machine (SVM) have distinct strengths and limitations, making them suitable for different scenarios.

Logistic Regression offers a transparent and computationally efficient approach, making it suitable for baseline disease classification. MLP, with its neural network architecture, can model highly nonlinear and complex relationships but requires significant computational resources and careful regularization. Random Forest provides a balance of robustness and accuracy, making it a reliable choice for aquaculture environments where sensor data may contain noise. SVM excels in classification tasks involving high-dimensional or limited datasets, though it may become computationally intensive when scaled. Collectively, these models provide

complementary capabilities and a hybrid or ensemble approach may yield optimal performance in real-world aquaculture disease prediction systems.

The proposed system utilizes a DC-powered pH strip sensor interfaced with an Arduino microcontroller to record voltage measurements corresponding to the acidity or alkalinity of the solution under test. The voltage output of the sensor varies with pH levels, necessitating precise acquisition to capture even subtle changes. To enhance accuracy, a 16-bit ADC (ADS1115) is deployed. This module enables the processor to interpret weak analog signals by converting them into high-resolution digital values. The increased resolution ensures the reliable detection of minute voltage variations, which directly correlate with small pH fluctuations. The ADS1115 is particularly valuable in scenarios requiring high precision, such as scientific experiments and Internet of Things (IoT)-based environmental monitoring. Once the analog readings are digitized, the values are transmitted to a database where a structured dataset is generated. This dataset consists of voltage attributes mapped to corresponding pH conditions, forming the foundation for further Machine Learning (ML) based analysis to develop predictive capabilities, the acquired dataset is divided into training and testing subsets.

The training set is utilized to allow the model to learn patterns and relationships, while the testing set is reserved for performance evaluation. During model initialization, a suitable algorithm is selected and initial parameters are defined. The training phase involves iterative adjustment of model parameters to minimize prediction errors. Once trained, the model undergoes prediction and evaluation, where it generates outputs for the test set. Performance metrics such as accuracy, precision and reliability are employed to assess its effectiveness. Additionally, hyperparameter tuning is conducted to optimize the model configuration, ensuring robust prediction performance. The proposed architecture was implemented in real-world aquaculture fields and analyzed for various parametric values. Sensor data acquisition was coupled with a self-cleaning mechanism to ensure accuracy and minimize manual intervention. In addition, machine learning algorithms were integrated to detect anomalies and sensor failures, thereby enhancing the system's reliability. Key water quality parameters are temperature, turbidity, pH and TDS were continuously monitored to capture the precise real-time conditions of shrimp ponds. Monitoring these parameters is essential for detecting and preventing diseases that adversely affect shrimp populations. Raw data collected from sensors was calibrated on the server side to improve the overall efficiency and accuracy of the system.

The Disease–Water Quality Matrix is given in Table 5. This table guides for the feature selection for machine learning models, ensuring that the most influential parameters are used for prediction. It also provides risk labels for historical data, enabling supervised learning and validation of model accuracy. During deployment, real-time sensor readings are compared against matrix thresholds to generate early warning

TABLE 5. Disease influenced on water quality parameters [38].

Parameter	Range / Level	Disease Risk Level	Diseases Commonly Influenced
pH	7.5 – 8.5	Very Low	No disease induction
	7.0 – 7.4	Low	Early-stage Vibriosis
	6.5 – 6.9	Medium	Vibriosis, early AHPND (EMS)
	6.0 – 6.4	High	AHPND (EMS), WSSV
	< 6.0	Very High	Mass mortality, systemic infections
DO (mg/L)	> 6.0	Very Low	Normal health
	4.0 – 6.0	Low	Gill stress, reduced growth
	3.0 – 3.9	Medium	Bacterial gill disease
	2.0 – 2.9	High	Septicemia, gill necrosis
	< 2.0	Very High	Acute mortality events
Temperature (°C)	26 – 30	Very Low	Optimal condition
	30 – 32	Low	Vibriosis
	32 – 34	Medium	WSSV, bacterial infections
	> 34	High	Viral outbreaks, immune suppression
	Sudden fluctuation (>3 °C/day)	Very High	Disease outbreaks, mortality

alerts. The matrix can be updated to incorporate site-specific or seasonal variations, allowing models to adapt and maintain reliable disease prediction. By linking environmental parameters to disease outcomes, the matrix provides a scientifically grounded foundation for predictive analytics and proactive aquaculture management.

The monitoring and management of shrimp health remain crucial for ensuring sustainable aquaculture practices. To address this need, the proposed system is equipped with precise sensors and advanced processing techniques to generate timely alerts based on deviations in water quality.

These alerts act as an early warning system to prevent the outbreak of diseases and safeguard shrimp populations. Representative measurements recorded during implementation includes the system incorporating the environmental monitoring disease alert mechanism targeting common shrimp diseases.

In field validation, the system successfully generated an early alert for White Spot Disease, demonstrating its potential to mitigate disease-related losses in aquaculture. To improve farmer accessibility, an Android-based application named “Smart Agri” was developed and deployed. The application features voice assistance and a multi-lingual interface in Tamil and Telugu, enabling farmers to interact with the system in their native language. Through the Smart Agri application, users can seamlessly view real-time sensor readings water temperature, pH, dissolved oxygen, turbidity and TDS in an intuitive and comprehensible format. The system also provides a centralized dashboard, offering a comprehensive overview of aquaculture pond health across

different parameters. This integration of IoT-enabled sensing, machine learning and user-centric mobile applications offers a complete framework for sustainable shrimp farming. By providing real-time insights and disease prediction alerts in regional languages, the system empowers farmers with actionable information to ensure the health, productivity and sustainability of aquaculture operations.

In the proposed system the sensors measuring pH, dissolved oxygen and temperature were deployed on IP68-rated AquaBuoy platforms to maintain consistent immersion and minimize variability due to water level fluctuations. Calibration using standard reference solutions was performed regularly to ensure long-term measurement accuracy. Real-time data were continuously acquired and preprocessed locally to reduce noise, normalize features, and prepare inputs for TinyML models running on edge devices. These models perform on-site inference for early anomaly detection, stress identification, and disease prediction, operating efficiently under low-power constraints and without relying on continuous cloud connectivity. The water quality disease relationship was systematically modeled using physiological thresholds, statistical correlations, feature importance analysis, and time-series modeling to capture delayed and cumulative effects of environmental stress. Continuous model updates allow adaptation to site-specific and seasonal variations, ensuring reliable, real-time, and actionable insights for precision aquaculture management.

This integrated methodology combines continuous real-time monitoring, edge-based TinyML analytics and water quality–disease modeling, providing a robust foundation for precision aquaculture. By addressing both environmental variability and site-specific conditions, the framework ensures accurate, actionable insights into pond health and potential disease risks. The next section presents the experimental validation, analysis of model predictions and interpretation of water quality–disease relationships, highlighting the practical effectiveness and scientific contributions of the proposed system in comparison to existing approaches.

System robustness was addressed at both the hardware and software levels to ensure reliable long-term operation in aquaculture environments. At the hardware level, IP68-rated industrial sensors with built-in signal filtering were employed to mitigate the effects of moisture, biofouling, and environmental noise, thereby ensuring stable and repeatable measurements under varying pond conditions. At the software level, a Real-Time Operating System (RTOS) was implemented with task-level separation for sensing, signal conditioning, communication, and data logging. This deterministic task scheduling minimizes data loss, prevents timing conflicts, and ensures reliable signal acquisition and transmission even under continuous operation and variable network conditions.

The effectiveness of the proposed framework is examined in the subsequent results and discussion section through systematic experimental evaluation under real-world aquaculture conditions.

IV. RESULTS AND DISCUSSIONS

The measurements from the Aquabouy with sensor suite (pH, DO, ORP, temperature, turbidity and salinity) are first time-stamped and annotated with sensor metadata (sensor ID, pond ID, Global Positioning System(GPS) location, battery status) at the edge (ESP32 + ADS1115).

In the cloud, an ETL pipeline ingests the incoming stream into a versioned data lake where both raw and cleaned records are retained to ensure reproducibility. Feature engineering generates derived attributes (rolling means, rate-of-change, diurnal aggregates and lagged features) and integrates external covariates where available (weather, tide). For supervised tasks, datasets are labeled using field observations and expert annotations (disease events, calibration benchmarks).

The curated dataset is split into training, validation and test partitions (e.g., 70/15/15) with stratification to preserve class balance. Data are scaled (min-max or standardization) and persisted in model ready formats. Finally, a model registry tracks experiments, hyperparameters and evaluation metrics like accuracy, coefficient of determination (R^2), Root Mean Square Error (RMSE), and Mean Absolute Error (MAE), enabling the systematic retraining and safe deployment of compressed models for TinyML inference at the edge. Security, access control, and audit logging are maintained throughout the pipeline. These algorithms, including logistic regression, MLP, Random Forest and SVM, were employed to analyze the dataset and predict outcomes based on the observed water quality parameters. Through this testing process, the efficacy and performance of each algorithm were assessed, providing valuable insights into their suitability for predicting water quality attributes and facilitating informed decision-making in aquaculture management. The dataset was subjected to testing using various machine learning algorithms. pH levels predominantly cluster around neutrality, while turbidity, organic carbon, conductivity, sulfate, chloramines, solids and hardness exhibit specific concentration ranges. These findings provide insights into the common ranges and prevalent levels of these parameters within the dataset. Understanding these distributions is crucial for assessing the overall water quality and identifying any potential anomalies or trends that may impact aquatic ecosystems or fish health.

Fig. 5(a). illustrates the distribution of pH values, revealing a normal distribution with a prominent peak at pH 7, indicating that neutral pH levels are most frequent within the dataset. Fig. 5(b) presents the turbidity distribution, where the highest frequencies are concentrated between 3.5 and 4.5 NTU, reflecting the most common turbidity range observed. Fig. 5(c) displays the distribution of organic carbon, with maximum frequencies clustered between 12 and 16 mg/L, highlighting the dominant concentration range. Fig. 5(d) depicts the conductivity distribution, where peak frequencies are observed between 350 and 450 $\mu\text{S}/\text{cm}$, indicating a prevalent conductivity range. Fig. 5(e) shows the

sulfate distribution, with the highest frequency occurring near 335 mg/L, suggesting a common sulfate level in the dataset.

Fig. 5(f). illustrates the distribution of chloramines, where most frequencies are concentrated between 6 and 8 mg/L, highlighting a typical range for this parameter. Finally, Fig. 5(g). and Fig. 5(h). present the distributions of solids and hardness, respectively. Solids show the highest frequency between 15,000 and 25,000 mg/L, indicating the dominant range, while hardness values are predominantly concentrated between 180 and 220 mg/L, representing the most common hardness level recorded. Together, these distributions provide valuable insights into the typical ranges of key water quality parameters within the dataset. The distribution plots in Fig. 5(a–h). collectively reveal that the water quality parameters across the dataset are generally within optimal ranges for aquaculture operations. The normal distribution of pH with a peak at neutrality (pH 7) indicates a stable aquatic environment suitable for shrimp and fish health. The turbidity range (3.5–4.5 NTU) suggests relatively clear water with minimal suspended solids, reducing the risk of gill irritation and stress to cultured species. The concentration of organic carbon between 12–16 mg/L and conductivity around 350–450 $\mu\text{S}/\text{cm}$ indicates moderate nutrient levels, supporting microbial activity without causing eutrophication. Similarly, sulfate and chloramine levels cluster around safe thresholds, implying controlled chemical composition in the water. The solids range (15,000–25,000 mg/L) and hardness levels (180–220 mg/L) fall within acceptable limits for Brackishwater aquaculture, supporting ionic balance and shell formation in shrimp. Overall, these insights confirm that the dataset reflects a well-maintained culture environment with balanced physicochemical parameters. This also validates the reliability of the dataset for training machine learning models, as the parameters capture realistic aquaculture conditions without being overly skewed by anomalies or extreme events.

Figure 6(a). shows the correlation matrix, revealing subtle interactions between pH and key water quality parameters. pH exhibits a weak negative correlation with turbidity and a weak positive correlation with organic carbon, while correlations with conductivity, sulfate, chloramines, total solids, and hardness are negligible, indicating limited influence. These findings provide insights into water chemistry dynamics. Figure 6(b). depicts the effect of training set size on Random Forest (RF) accuracy. Accuracy improves with larger training sets, reflecting better learning and generalization, but plateaus beyond a certain size, suggesting diminishing returns and enabling identification of the optimal dataset size. Figure 6(c). compares model accuracies, with RF achieving the highest performance at 98.35%. Its superior accuracy is attributed to the ensemble approach, which integrates multiple decision trees to enhance predictive reliability and reduce overfitting. The dataset was used to predict voltage across pH levels 5–12. The Random Forest (RF) algorithm outperformed all tested models, achieving an RMSE of 0.01 V and

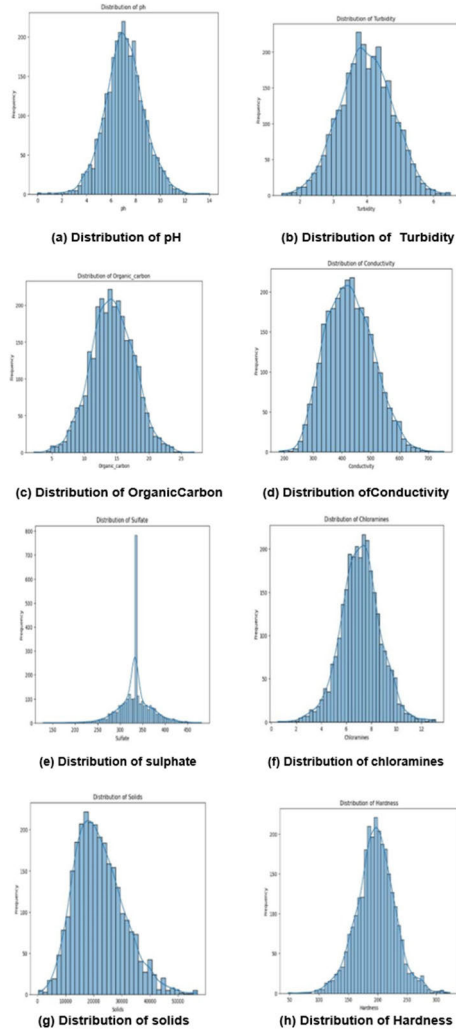
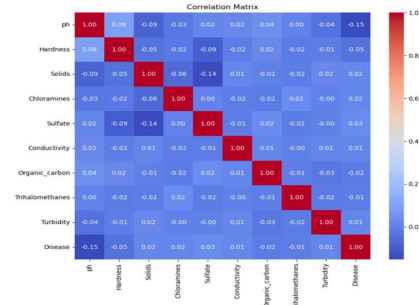


FIGURE 5. (a-h). Distribution of various parameters in the dataset.

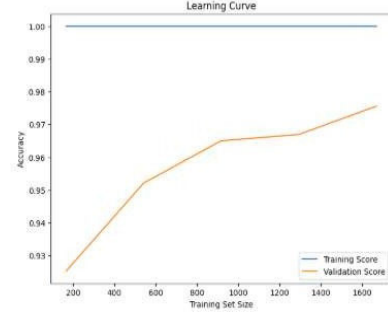
R^2 of 0.99. Its ability to account for environmental variations and sensor drift enabled consistent and accurate pH readings.

Figure 7. shows the strong agreement between actual and predicted voltages, demonstrating high prediction accuracy. calibration, reducing manual adjustments and enabling precise real-time monitoring in aquaculture systems. Voltage measurements exhibit well defined noise characteristics, with peak-to-peak amplitudes of ± 0.002 – ± 0.003 V, slightly higher at pH 9–12. Noise shows a high-frequency, Gaussian-like distribution, while lower pH ranges (5–8) maintain stable baselines. Over 800 s, the noise profile remains stable, and the signal-to-noise ratio allows accurate discrimination between adjacent pH levels.

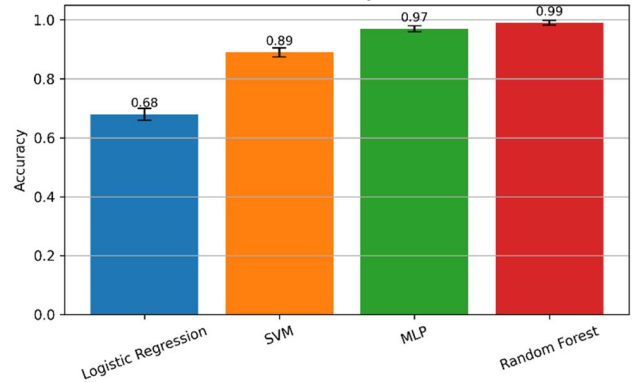
Figure 8. illustrates that voltage remains relatively constant over time for each pH, with higher pH corresponding to lower (more negative) voltages. Error bars indicate variability in predicted voltages for each pH level. Most pH levels, especially mid-range values (pH 7–10), show negligible variability, reflecting high model confidence. Slightly larger error bars at the extremes (pH 5 and 12) suggest increased



(a). Correlation Matrix



(b). Size of the training set Vs accuracy



(c). Accuracy of the ML algorithms

FIGURE 6. Correlation and performance analysis of ML models.

prediction uncertainty, likely due to fewer data points or greater sensor noise. Overall, predicted voltages decrease consistently with increasing pH, confirming the model's reliability and the sensor's stability.

Figure 9 (a). illustrates the relationship between predicted voltage for various pH values, showing a clear negative correlation of higher pH levels correspond to more negative voltages. The scatter plot reveals a consistent trend, confirming that the prediction model accurately captures the pH voltage relationship. Predicted voltages range from 0.14 V to -0.01 V across pH 5–12, with four representative points at pH 5, 7, 10 and 12, indicating an approximately linear relationship and a sensitivity of ~ 0.02 – 0.03 V per pH unit. The consistent spacing between voltage measurements suggests uniform sensor sensitivity over the pH range. Figure 9 (b). presents the temporal stability of voltage measurements for different pH levels.

Voltages remain largely constant over time, demonstrating minimal drift and consistent sensor performance. Higher pH

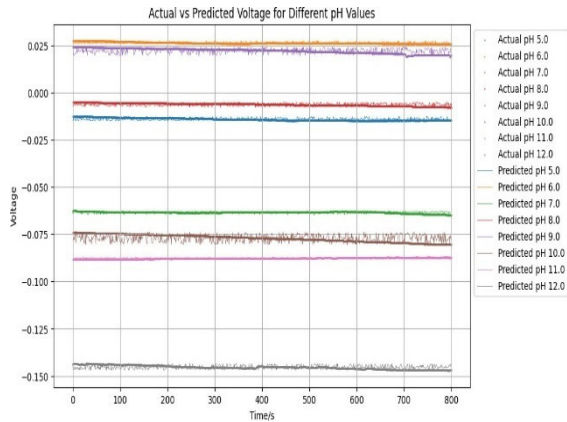


FIGURE 7. Actual Vs predicted voltage for various pH values.

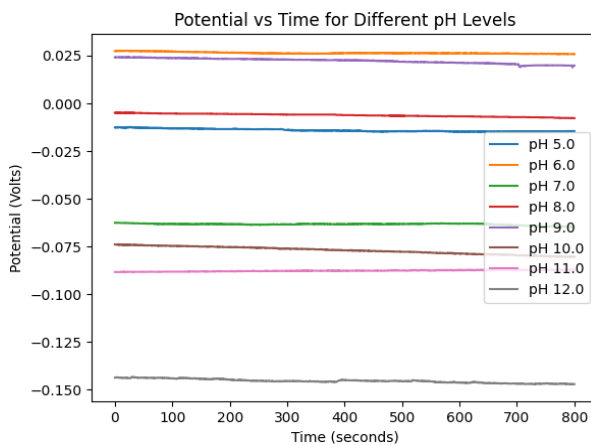


FIGURE 8. Potential Vs time for different pH levels.

values (e.g., pH 12) correspond to more negative voltages, while lower pH values (e.g., pH 5) exhibit less negative or slightly positive voltages. Minor fluctuations are attributed to environmental factors or inherent sensor noise, but overall, the data indicate stable and reliable voltage readings for accurate pH discrimination.

The predictive model was evaluated using time-series analysis of DO, ORP, and pH over 100-time steps in Figure 10. Actual values (dashed lines) closely follow predicted smoothed curves, confirming the model's ability to capture trends and fluctuations in water quality. Minor deviations indicate adaptability to changing conditions, supporting its reliability for real-time monitoring.

Training and validation loss curves Figure 11. shows the exponential decline over 20 epochs with minimal fluctuations, demonstrating stable optimization and resistance to overfitting. Time-series plots of DO, ORP, and pH further validate model performance, with DO predictions capturing dynamic ranges upto 55–60 mg/L. Errors of ± 5 mg/L during peak DO events are attributed to sensor noise and limited representation of extreme variations. A feature correlation heatmap highlights key predictors while reducing redundancy, improving both computational efficiency and forecast accuracy.

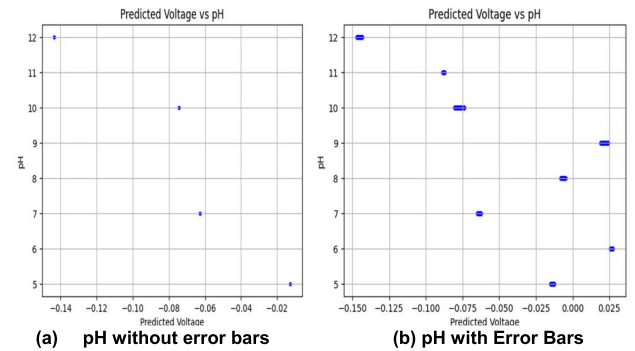


FIGURE 9. Predicted voltage Vs pH without & with error bars.

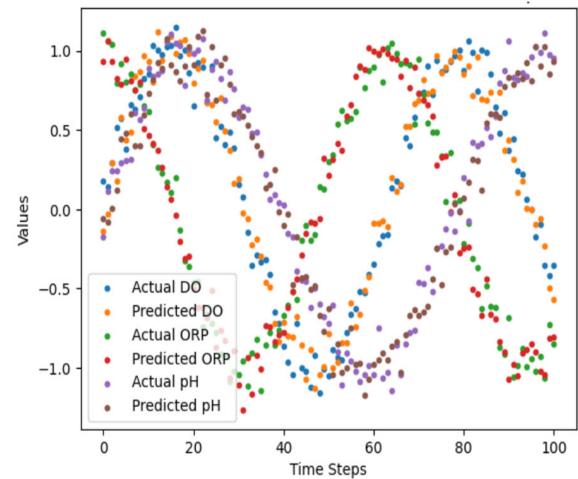


FIGURE 10. Actual Vs. Predicted DO, ORP and pH over time.

Residual plots in Figure 12. display near-normal error distribution, with minor systematic bias in ORP, while DO and pH remain well-balanced. Overall, the results confirm that the proposed model is efficient, reliable, and suitable for real-time water quality monitoring.

Performance metrics (RMSE, MAE, R^2) confirm the model's robustness, with R^2 approaching 1, indicating strong predictive accuracy and minimal deviation. Specifically, for DO: Figure 13(a) and pH: Figure 13(b) low RMSE and MAE values combined with $R^2 \approx 0.9$ demonstrate precise forecasting capability.

Figure 14. illustrates the data acquisition information through the MQTT protocol and presents the water buoy dashboard developed for shrimp disease alerts. The monitoring and management of shrimp health in aquaculture are critical to ensuring both sustainability and productivity. To address this challenge, the proposed system has been designed to generate timely alerts by continuously measuring key water quality parameters. These measurements serve as early indicators for predicting and preventing disease outbreaks that can severely impact shrimp populations.

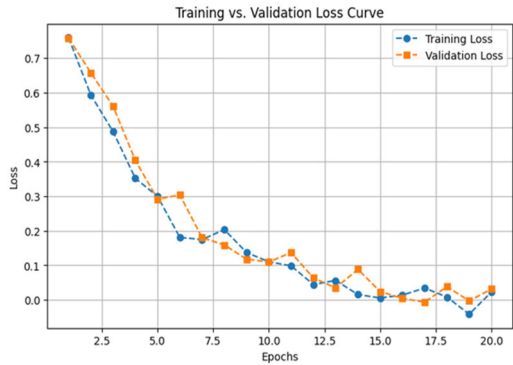


FIGURE 11. Training Vs. validation loss curve.

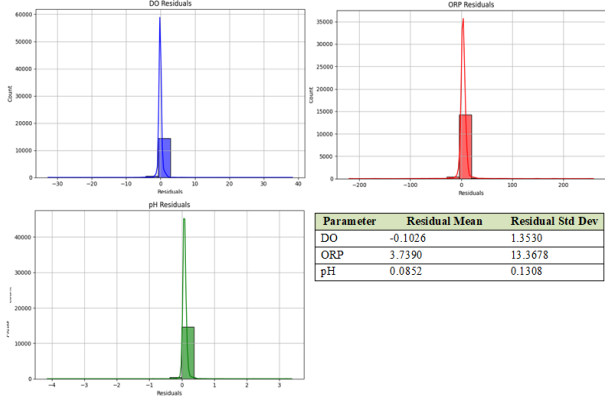


FIGURE 12. Residual plot for DO, ORP and pH.

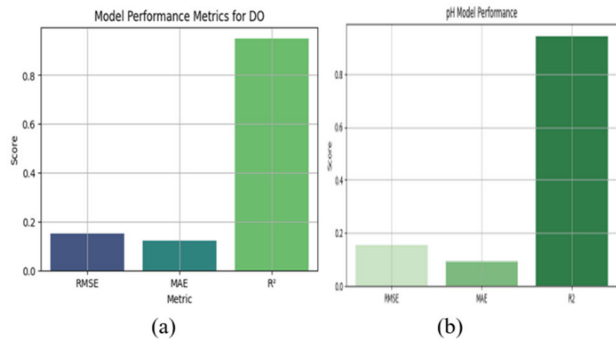


FIGURE 13. Model performance for DO & pH.

The monitoring and management of shrimp health in aquaculture settings is crucial for ensuring a successful and sustainable operation. To address this concern, a system has been designed to provide timely alerts based on the measurement of key water quality parameters. These measurements play a pivotal role in predicting and preventing the outbreak of diseases that can negatively impact shrimp populations. The proposed monitoring system, equipped with precise sensors and advanced technology, has been diligently measuring several crucial parameters in our shrimp farming environment.

These measurements serve as an early warning system to safeguard the health of our shrimp population.



IoT-Based Smart Buoy Monitoring and Disease Alert Dashboard

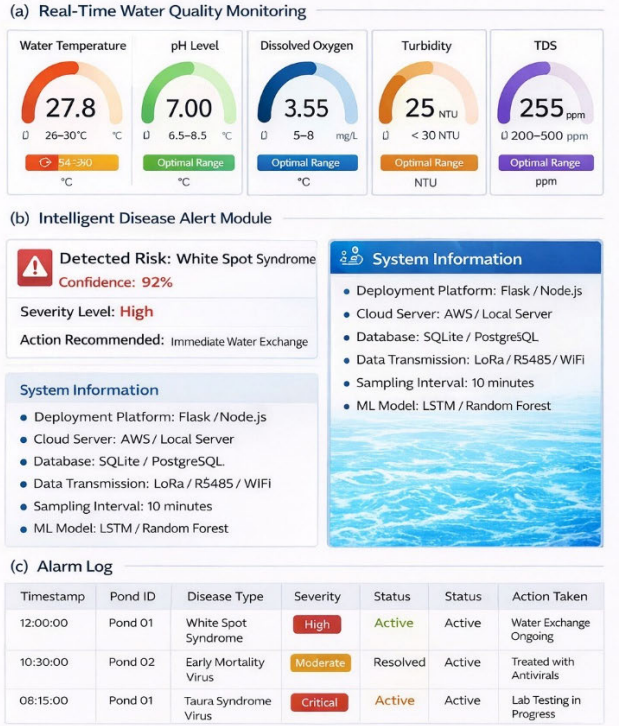


FIGURE 14. Web app deployment and dashboard design.

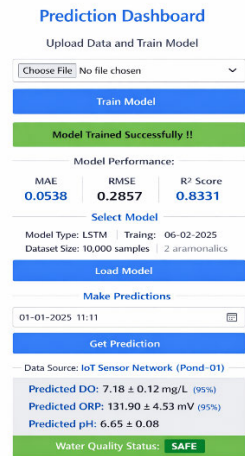


FIGURE 15. Prediction dashboard.

a) Water Temperature: 27.81 °C Optimal water temperature is crucial for the well-being of shrimp. At 27.81 °C, system ensures that the temperature remains within the desired range for shrimp farming.

b) pH Level: 7.00 A neutral pH of 7.00 is essential for maintaining a healthy aquatic environment. This system constantly monitors and regulates the pH level to ensure that it is ideal for shrimp growth.



IoT-Enabled Smart Pond Monitoring and Prediction Dashboard

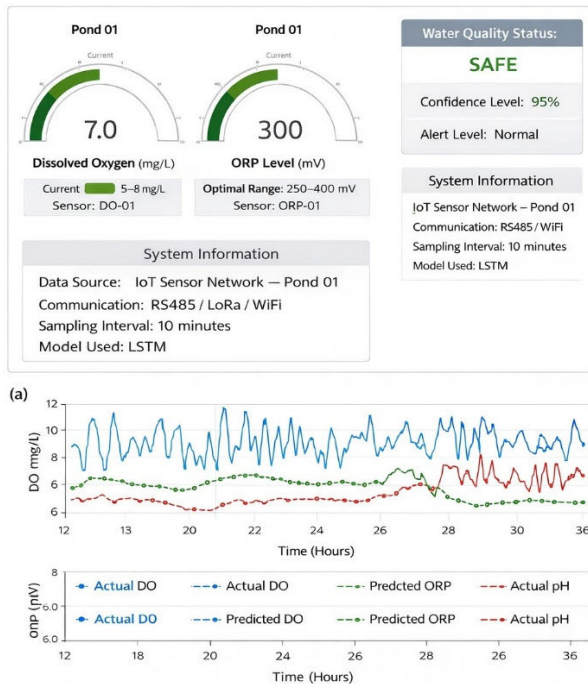


FIGURE 16. Pond dashboard.

c) Dissolved Oxygen: 3.55 mg/L Dissolved oxygen is a critical factor in aquatic ecosystems. With a reading of 3.55 mg/L, system ensures that there is an adequate oxygen supply to support the respiration of the shrimp.

d) Turbidity: 25 NTU Turbidity, or water clarity, plays a significant role in the overall health of the shrimp. A turbidity level of 25 NTU is well within the recommended range, providing an environment in which shrimp can thrive.

e) Total Dissolved Solids: 255 ppm The Total Dissolved Solids (TDS) in water can affect shrimp health. The proposed system has maintained a TDS level of 255 ppm, which is conducive to shrimp farming.

The system integrates multiple high-precision sensors and advanced IoT technology for real-time monitoring of critical parameters, including pH, DO, ORP, turbidity, and Total Dissolved Solids (TDS). In addition to continuous monitoring, the architecture incorporates an alert mechanism capable of detecting four major shrimp diseases. During testing, the system successfully generated an early alert for White Spot Disease, thereby demonstrating its practical utility in aquaculture disease prevention.

A centralized dashboard has been developed to visualize overall pond conditions, with three key metrics like pH, DO, and ORP highlighted at the edge, as shown in Figure 15. The real-time monitoring system empowers aquaculture managers to make proactive decisions by examining variations in pH, DO and ORP levels. Such insights help in

maintaining stable aquatic conditions, increasing shrimp yield, and reducing mortality risks. The predictive capabilities of the system are further enhanced through machine learning models. By applying regression algorithms and Support Vector Machines (SVMs), the system calibrates sensor readings to improve the accuracy of pH measurements. This ensures precise and dependable monitoring of water quality parameters.

Semi-circular gauge meters in Figure 16. provide intuitive visualization of water quality, enabling rapid assessment of pond health. The system also supports multi-pond monitoring, allowing users to select and track specific pond environments in real time, as indicated by labels such as Pond ID: Pond_001.

The graphical dashboard provides trend visualizations, where ORP graphs reveal dynamic oxidation changes, and pH and DO trends highlight variations in oxygen availability and acidity/alkalinity. These visual insights enable aquaculture stakeholders to maintain optimal environmental conditions, thereby safeguarding shrimp health and supporting sustainable aquaculture practices.

Deployment challenges associated with implementing the proposed AI-IoT framework in shrimp aquaculture farms include limited power availability in remote pond locations, connectivity constraints in rural areas, maintaining long-term sensor accuracy, ensuring sensor durability under harsh environmental conditions, and securing reliable data transmission against potential losses or corruption; these challenges are addressed through low-power sensing hardware, energy-efficient communication protocols, edge-level data processing, automated calibration with periodic maintenance, robust sensor enclosures, and reliable IoT communication strategies to enable scalable, accurate, and resilient field deployment.

V. CONCLUSION

This study introduces a robust AI- and IoT-enabled framework that integrates real-time sensing, predictive analytics and automated pH sensor calibration for precision aquaculture management. The Aquabouy-based monitoring system successfully measures key water quality parameters such as pH, DO and ORP through a distributed IoT network with cloud-based data storage. Comparative machine learning analysis highlights Random Forest as the top performer for disease prediction with an accuracy of 98.350.99, RMSE = 0.01 V), significantly reducing manual calibration efforts. LSTM models effectively capture temporal dependencies in time-series data, achieving up to 94 accuracy in predicting water quality trends. The integration of a centralized web dashboard and multilingual Android application enhances accessibility, providing real-time visualization, decision support and automated alerts to farmers, thereby minimizing shrimp mortality risks and improving operational efficiency.

Future work will focus on expanding the dataset across multiple farms and seasonal variations to improve model robustness and generalizability. Advanced architectures such as hybrid CNN-LSTM and attention-based deep learning

models will be explored for higher accuracy in forecasting complex water quality dynamics. Edge AI deployment will be considered to enable offline predictions and reduce cloud dependency, improving scalability in remote areas. Additionally, integrating blockchain technology can provide end-to-end traceability in shrimp supply chains, linking sensor data to transparent records. Automated control systems for aerators and dosing units will be developed to achieve closed-loop water management, ensuring sustainable and data-driven aquaculture practices.

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